

Structurally-Calibrated Functional Attribution for Audit Prioritization in Knowledge-Intensive Reasoning

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ABSTRACT

Information and knowledge systems increasingly expose reasoning traces, process annotations, retrieval checks, graph nodes, entity bindings, and verification records as intermediate knowledge artifacts. These artifacts can support audit and curation, but they are often stored as heterogeneous signals rather than records that preserve dependency, uncertainty, and maintenance context under fixed review budgets. Existing process-annotation, scalar attribution, and graph-salience methods provide useful local signals but not a reusable representation for deciding what an artifact is, why it matters structurally, and how it should be audited. We introduce Structurally-Calibrated Functional Attribution (SC-FMA), a knowledge artifact representation framework for fixed-budget audit-record construction. SC-FMA represents artifacts as dependency-aware audit graphs and uses the Structurally-Calibrated Utility (SCU) objective to construct records that preserve annotation or utility fidelity while exposing structural dependency, redundancy, bottleneck, audit-reason, and maintenance-action fields. Spearman, NDCG, and retrieval readouts are used as representation-fidelity and audit-coverage diagnostics. On a locked process-annotation distribution derived from PRM800K with 4,417 samples and 34,219 labeled artifacts, SC-FMA Ridge preserves annotation-order fidelity inside the audit record (Spearman $\rho = 0.604$ versus 0.611 for $\mathbf{w}_{\text{struct}}$) while exposing decomposed structural and maintenance fields. A Countries-KG diagnostic graph shows that ontology-derived edges change structural-dependency representation, and rule-derived audit-target retrieval improves budget-constrained coverage from 0.235 to 0.699 over a scalar-only view, with circularity bounded by target type. SC-FMA provides an information-structuring layer for auditable knowledge maintenance and curation.

1. Introduction

Knowledge-intensive information systems increasingly expose intermediate artifacts during reasoning, retrieval, annotation, validation, and update. These artifacts include reasoning traces, graph nodes, entity bindings, verification records, rule-like operations, retrieval checks, and process annotations. Once exposed, they are no longer only transient computational by-products: they become knowledge artifacts that must be represented, organized, maintained, curated, and reused. The practical challenge is that these artifacts are heterogeneous, uncertain, and often more numerous than the audit capacity available to curators, while maintenance decisions must still be made under fixed review budgets.

Existing methods address only parts of this information-structuring problem. Process-annotation methods attach scalar quality signals to intermediate artifacts, but they do not convert those annotations into structured maintenance records. Local utility-signal methods estimate artifact influence, but they do not distinguish whether an artifact should be inspected because it provides local evidence, supports a dependency, duplicates existing support, or exposes a downstream bottleneck. Graph-based salience methods can identify structurally prominent nodes, but they do not integrate annotation signals, dependency structure, and curation actions into a reusable audit representation. Consequently, these methods can yield audit priorities, but they cannot specify what the artifact is, why it matters in the knowledge structure, and what maintenance action a fixed-budget audit should take.

This paper introduces Structurally-Calibrated Functional Attribution (SC-FMA) as a knowledge representation transformation layer for this setting. SC-FMA transforms observable intermediate knowledge artifacts into structured audit records by representing their dependencies, calibrating supplied annotation or utility signals, and attaching explicit audit reasons and maintenance actions. The audit-priority field is therefore a downstream use of the representation, not the identity of the method. The central contribution is to treat audit under limited review capacity

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as an information representation problem: SC-FMA converts raw traces, graph relations, and process annotations into dependency-aware records that can be inspected, queried, curated, and reused.

The paper makes three contributions. First, it formulates fixed-budget audit-record construction for structured knowledge artifacts, where intermediate records must be selected for review under limited curation resources. Second, it introduces SC-FMA and the SCU objective, which combine annotation fidelity, structural dependency, redundancy, and bottleneck roles into decomposed audit records. Third, it evaluates the representation on a PRM800K process-annotation dataset, a Countries-KG knowledge representation diagnostic graph, rule-derived audit-target retrieval, and controlled synthetic calibration, keeping deployment and downstream-training claims separate from the reported representation-level audit evidence.

The main findings follow the same knowledge-engineering workflow. On the PRM800K process-annotation dataset (4,417 samples and 34,219 labeled artifacts under a frozen hash split), $\mathbf{w}_{\text{struct}}$ is treated as a fidelity field for artifact records, and SC-FMA Ridge closely preserves annotation-order consistency (Spearman $\rho = 0.604$ versus 0.611) while exposing decomposition fields for audit; this is a preservation-with-decomposition result, not a predictive-improvement result over $\mathbf{w}_{\text{struct}}$. The Countries-KG diagnostic graph shows that typed ontology edges can change structural-dependency representation at the graph-representation interface. Rule-derived audit-target retrieval shows that SC-FMA decomposition exposes bottleneck, redundancy, weak-anchor, and structural-overcorrection targets more fully than a scalar-only view under the same fixed audit budget, but these targets are automatic rules rather than human adjudications and three target families share partial circularity with the decomposition fields. Controlled synthetic calibration and ablation explain when structural fields affect representation-field assignment and when they mainly provide regularization and audit-record decomposition.

The remainder of this paper is organized as follows. Section 2 reviews attribution, step-level annotation signals, graph-based knowledge engineering, and audit methods. Section 3 presents SC-FMA, the SCU objective, and its theoretical properties. Section 4 reports knowledge-engineering task validation from real annotation evidence to knowledge-graph representation and controlled calibration. Section 5 interprets the method in the knowledge lifecycle, and Section 6 concludes.

Figure 1 gives the high-level knowledge-engineering workflow. SC-FMA sits between raw knowledge artifacts and downstream maintenance actions: it represents artifacts as dependency graphs, calibrates annotation signals with SCU, and returns knowledge audit records with explicit structural roles and maintenance actions.

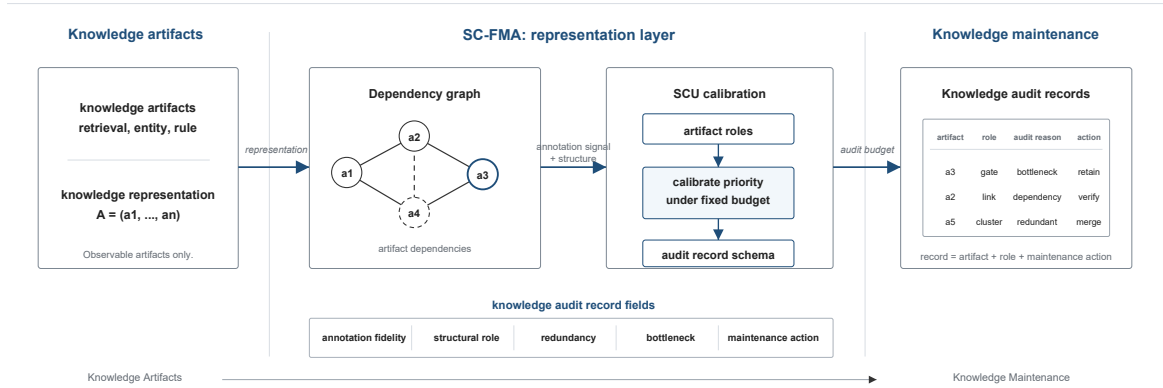


Figure 1: Knowledge engineering workflow for SC-FMA. The figure should be read as a transformation from Knowledge Artifact $\rightarrow$ Representation $\rightarrow$ Graph Structure $\rightarrow$ SCU $\rightarrow$ Audit Record $\rightarrow$ Knowledge Curation. Raw reasoning traces, graph nodes, and process annotations are first organized as auditable knowledge artifacts, then represented through dependency-aware graph structure. SCU constructs structured audit records by preserving annotation or utility fidelity while adding structural dependency, redundancy, bottleneck, audit-reason, and maintenance-action fields. The resulting records support fixed-budget knowledge maintenance and curation; the audit-priority field is only a downstream use of the audit representation.

2. Related Work

2.1. Attribution and Explanation

Attribution methods ask which parts of an input, computation, or structure matter for a decision. Integrated Gradients attributes predictions to input features along a path from a baseline [1]; SHAP unifies additive feature-attribution methods [2]; and LIME explains individual predictions through local interpretable surrogates [3]. Rationale-oriented work such as ERASER evaluates extractive spans rather than only continuous attribution values [4].

Structure-aware attribution adds another layer. Network studies show that importance can be non-local, redundant, or concentrated around bottleneck nodes [5, 6]. For graph models, GNNExplainer selects compact subgraphs that explain predictions [7]. User-centered explanation work further shows that an explanation is useful only when it matches the reviewer’s task [8].

These lines motivate SC-FMA but leave a gap for audit triage. Saliency identifies important evidence, yet it does not specify whether the reviewer should inspect local support, consolidate duplicated evidence, protect a downstream dependency, or repair a weak upstream signal. Entailment-tree methods move explanation toward intermediate justifications [9], and counterfactual methods move it toward contrastive evidence [10]. SC-FMA keeps these audit roles separate.

2.2. Process Supervision and Process Reward Models

Process supervision assigns annotation signals to intermediate artifacts rather than only final answers. The PRM800K process-annotation dataset showed that process reward models trained on per-artifact human labels can provide stronger verifier-guided best-of- N search behavior than outcome reward models [11]. The GSM8K verifier setup made this training route concrete [12], and process-based feedback reduced reasoning errors even when outcome success appeared saturated [13]. These studies provide the kind of artifact-resolved fidelity signal that SC-FMA can calibrate.

A parallel line makes verification, critique, and revision themselves available as annotated process units. Chain-of-thought prompting elicits intermediate reasoning [14], and self-consistency aggregates multiple sampled chains into a more reliable signal [15]. Tree of Thoughts extends search from linear chains to deliberate tree-structured exploration [16], while ReAct interleaves reasoning and acting for verification [17]. On the revision side, Reflexion [18] and Self-Refine [19] introduced reflective agents that critique and revise their own outputs, turning revision steps into an additional supervision target.

Recent process-diagnostic resources also show that step-level signals need distributional checks. Math-Shepherd studies automatic step verification without human annotations [20], and ProcessBench evaluates process-error identification across models [21]. These resources help assess scalar step signals. SC-FMA treats such signals as inputs to structural calibration rather than as final explanations.

2.3. Graph-Based Knowledge Engineering and Audit Methods

Graph-based knowledge engineering shows how structured intermediate states can be made inspectable. Knowledge-engineering methods have long treated representation, inference, and maintenance as coupled design problems [22]. Classic rule-based systems treated auditability as a design principle: MYCIN exposed certainty-factor chains that connected conclusions to rules [23], and DENDRAL exposed generate-and-test plans for chemical structure enumeration [24]. Cognitive architectures such as Soar and ACT-R also made internal states and operator applications available for step-level review [25, 26].

Modern knowledge-intensive pipelines shift the review focus from hand-built rules to learned workflows that connect language models with structured knowledge. Knowledge graphs provide a general representation substrate for entities, relations, constraints, and graph-level quality signals [27]. RAG exposes retrieved passages as evidence [28], and survey work documents the broader family of RAG variants [29]. GraphRAG builds entity graphs and pre-generates community summaries for global queries [30]. KG-RAG connects a pre-existing knowledge graph to LLM generation through chain-of-exploration reasoning [31]. Knowledge-enhanced language models and LLM-KG integration further expose entity links, triples, relation paths, and graph-quality signals as knowledge artifacts for audit [32–36]. Similar audit pressure appears in long-tail KG-augmented question answering and engineering-design RAG, where sparse factual support and heterogeneous evidence make exhaustive review unrealistic [37–39].

Across these settings, the challenge is no longer simply recovering intermediate evidence. Modern pipelines already expose passages, entity links, relation paths, and verification records. The harder question is how a reviewer should

allocate a limited curation budget across them and how the resulting records should support refinement, update, and reuse [40, 41]. SC-FMA connects to this knowledge-engineering audit tradition by converting exposed artifacts into graph-aware audit records.

2.4. Summary of Research Gap

Existing work supplies three ingredients: local attribution identifies salient evidence, process supervision supplies step-level annotation fidelity, and graph-based knowledge engineering exposes structural context. SC-FMA combines them into a representation layer that turns exposed knowledge artifacts into inspectable records for fixed-budget knowledge audit.

3. Method

3.1. Workflow Overview

Representation Transformation Statement. SC-FMA transforms raw intermediate reasoning artifacts into structured, queryable knowledge audit representations. The method should be read as a knowledge representation layer between exposed artifact streams and downstream maintenance actions, not as a standalone prioritization heuristic. SCU does not define a scalar decision rule, but defines a structured representation constraint system over knowledge artifacts.

The input is a set of observable knowledge artifacts such as retrieval checks, entity bindings, rule-like operations, verification records, or reflective annotations. These artifacts are first normalized into an auditable artifact sequence, converted into a graph representation with temporal and optional domain edges, calibrated by SCU against the supplied annotation or utility signal, and emitted as knowledge audit records. A downstream curator can then use those records to allocate a fixed audit budget while seeing both the audit-priority value and the maintenance reason for inspection.

3.2. SC-FMA as a Representation Layer

Unlike conventional scalar attribution methods, SC-FMA explicitly represents structural dependencies among intermediate knowledge artifacts and converts them into audit-oriented knowledge records. The method acts as a representation layer between raw artifacts and downstream maintenance actions: it does not replace the reasoning system, the retriever, the knowledge graph, or the annotator, but reorganizes their exposed outputs into records that preserve fidelity information while adding structural role and maintenance-action fields.

3.3. Formal Problem Definition

Input. SC-FMA takes an observable artifact sequence $T = (s_1, \dots, s_k)$, where each unit s_i may be a verification, reflection, retrieval, entity-binding, or rule-like operation available for audit. The sequence is converted into a graph $G = (V, E)$ whose nodes correspond to artifact units and whose edges encode temporal adjacency plus optional topical or domain-specific relations. The method also receives a utility or fidelity signal u over the same units. This signal may come from local utility estimation, a proxy representation signal, or a learned step-annotation model, and SC-FMA treats it as the fidelity signal to be preserved when it is informative.

Output and objective. The output is a set of calibrated knowledge artifact records. Each record contains an audit-priority field plus fidelity, structural-dependency, redundancy, bottleneck, audit-reason, and maintenance-action fields. The methodological object is the structured record rather than a priority list alone. The objective is to preserve useful utility signal, incorporate observable artifact structure, and attach each selected artifact to a concrete audit action.

3.4. SCU Objective and Audit Decomposition

The SCU objective is designed to produce an executable audit representation, not merely a scalar priority value. Its role is to preserve an informative fidelity signal while making the structural reason for each priority visible under a fixed budget. Dependency, redundancy, and bottleneck fields therefore do not serve as decorative metadata; they determine whether a selected knowledge artifact should be checked as local evidence, protected as a dependency, consolidated as repeated support, or retained because later artifacts depend on it.

For an artifact sequence with k auditable units, let $\tilde{c} \in \mathbb{R}^k$ denote the normalized utility or proxy-fidelity input. In the locked PRM800K process-annotation stage, $\mathbf{w}_{\text{struct}}$ denotes the frozen Ridge regressor used as this fidelity signal. In the binary-correctness diagnostic setting, $\tilde{c}$ is a coarse sequence-level local utility anchor: it records how an observed outcome changes under a documented intervention. Let $\tilde{\mathbf{n}}$ denote normalized structural necessity from graph perturbation diagnostics, $\mathbf{R}$ a redundancy matrix, and $\mathbf{b}$ a bottleneck indicator.

SC-FMA computes audit-allocation values $\mathbf{w}$ on the probability simplex

$$\mathcal{W} = \{\mathbf{w} \in \mathbb{R}_+^k : \sum_i w_i = 1\} \quad (1)$$

by minimizing the SCU objective:

$$L(\mathbf{w}) = \alpha \|\mathbf{w} - \tilde{\mathbf{c}}\|_2^2 + \beta \|\mathbf{w} - \tilde{\mathbf{n}}\|_2^2 + \gamma \mathbf{w}^\top \mathbf{R} \mathbf{w} - \delta \sum_i b_i \log(w_i). \quad (2)$$

Unless otherwise stated, $\alpha, \beta, \gamma, \delta \geq 0$ and the symmetric part of $\mathbf{R}$ is treated as positive semidefinite on the simplex tangent subspace. The four terms encode fidelity to the supplied input, structural alignment, redundancy regularization, and non-zero bottleneck regularization. The fidelity signal consistently provides the dominant optimization signal. The redundancy penalty and bottleneck log-barrier mainly act as regularization terms that encourage interpretable decomposition fields, although their quantitative contribution to annotation-order consistency in the audit record is modest under the current validation setting. Detailed notation, construction rules, algorithms, and proofs are provided in the supplementary material.

3.5. Artifact Representation

Each artifact sequence is represented as an ordered set of auditable knowledge artifacts. The current implementation supports units derived from structured reflection tags and free-text chain-of-thought segmentation. In this paper, segmentation is treated as an observed input rather than a solved NLP problem.

A node v_i corresponds to one artifact unit. Temporal edges connect adjacent units, while topical edges connect units whose TF-IDF similarity exceeds the configured threshold. The graph serves as a representation substrate for knowledge dependencies rather than a feature-extraction pipeline; TF-IDF is only the lightweight default backend for constructing lexical dependency cues when richer domain relations are unavailable. This temporal-plus-topical graph is the default SCU instantiation for text-only artifact sequences and gives the method an auditable structural record.

A supplementary graph-construction ablation replaces TF-IDF topical edges with Sentence-BERT embeddings (sentence-transformers/all-MiniLM-L6-v2, 384 dimensions). In that 800-trace fixture, the TF-IDF topical default (0.0515) did not exceed temporal-only edges (0.0596) on diagnostic structural-necessity consistency, while the embedding backend (0.0615) slightly improved the same diagnostic readout.

These differences are too small to treat topical edges as a representation-field driver. We therefore read the graph layer as an interchangeable *construction interface*, not as evidence that the TF-IDF topology adds annotation-order preservation value over the temporal chain.

The graph constructor is modular. Richer domain-specific backends — semantic embeddings, entity links, knowledge graphs, ontologies, rule-dependency DAGs, and data-lineage graphs — can replace the default without modifying the SCU objective. The Countries-KG representation study (Section 4.3) and the embedding ablation illustrate this interface. The resulting graph is a directed acyclic approximation of an observable artifact sequence and serves as a reproducible audit graph.

3.6. Structural Necessity, Redundancy, and Bottlenecks

Structural necessity measures how much graph support is lost when a node is perturbed under a documented graph-level operation. The implementation uses PRUNE, CASCADE, and BYPASS diagnostics. PRUNE removes direct node support. CASCADE propagates disruption to downstream neighbors. BYPASS asks whether alternative paths can substitute for the removed artifact unit. The final necessity input $\tilde{\mathbf{n}}$ is the normalized summary of these observable graph diagnostics.

The redundancy matrix $\mathbf{R}$ captures pairwise similarity between artifact units. High redundancy means that two units appear to serve overlapping functions in the observed sequence. The penalty $\mathbf{w}^\top \mathbf{R} \mathbf{w}$ discourages allocating large budget share to many mutually redundant units at once. In the current evidence package, this term is best interpreted as a regularization and decomposition term: it records a possible duplicated-audit representation pressure, but its direct contribution to annotation-order consistency is modest.

Bottleneck detection addresses the opposite diagnostic pattern. Some nodes are not redundant, and their downstream exposure can make them useful to inspect even if the supplied fidelity signal is modest. The log-barrier term prevents identified bottleneck nodes from being assigned near-zero audit allocation. In knowledge-base maintenance

terms, this keeps a single point of failure visible to the reviewer. The term enforces a conservative audit regularizer when observed structure marks the node as difficult to replace.

Figure 2 illustrates the local-utility–necessity mismatch. Figure 3 reports the three perturbation modes, and Figure 4 shows the resilience curves.

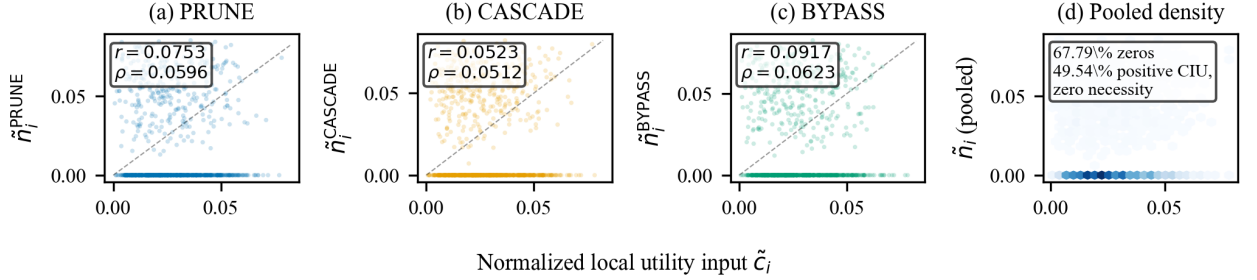


Figure 2: Structural diagnostic motivating calibration. Local utility and graph-level necessity are not interchangeable: many locally useful steps have low measured structural necessity, while some structurally exposed steps do not receive strong local utility signals. The shared x -axis $\tilde{c}_i$ denotes the normalized local utility input for step i . SC-FMA uses this mismatch as the reason for calibration.

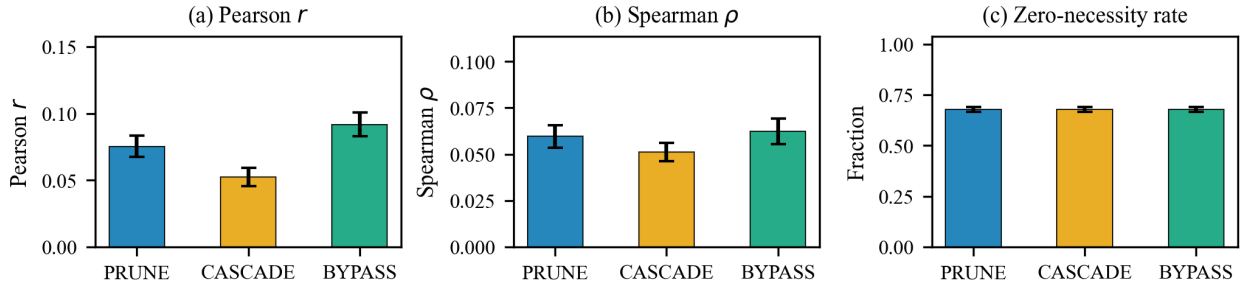


Figure 3: Structural diagnostic summary by perturbation mode. The three perturbation modes (PRUNE, CASCADE, BYPASS) are reported with Pearson correlation, Spearman annotation-order consistency, and zero-necessity fraction. Error bars indicate 95% bootstrap confidence intervals. The consistency across modes indicates that weak local-utility–necessity alignment is not an artifact of a single perturbation strategy.

Theorem 3.1 (SCU well-posedness). *Let $\mathbf{R}_s = (\mathbf{R} + \mathbf{R}^\top)/2$ and let $\mathcal{T} = \{\mathbf{z} \in \mathbb{R}^k : \mathbf{1}^\top \mathbf{z} = 0\}$ be the tangent subspace of the simplex. Assume $\alpha, \beta, \gamma, \delta \geq 0$ and $\mathbf{z}^\top \mathbf{R}_s \mathbf{z} \geq 0$ for every $\mathbf{z} \in \mathcal{T}$. Then the SCU objective in Eq. 2 is convex on the relative interior of the simplex in Eq. 1. If $\mathbf{z}^\top ((\alpha + \beta)\mathbf{I} + \gamma \mathbf{R}_s) \mathbf{z} > 0$ for every nonzero $\mathbf{z} \in \mathcal{T}$, the quadratic part is strictly convex on the feasible subspace and the minimizer is unique. If $\delta > 0$ and $b_i = 1$, any finite minimizer assigns $w_i > 0$ to that bottleneck node; the barrier therefore gives an interior guarantee for active bottleneck support. For non-redundant artifact-unit pairs with identical redundancy profiles and inputs ordered in the same direction, $\tilde{c}_i \geq \tilde{c}_j$ and $\tilde{n}_i \geq \tilde{n}_j$, the two-input allocation relation is monotone before redundancy-induced redistribution. This monotonicity is a diagnostic property, not a global ordering guarantee once redundancy profiles differ.*

We implement three variants of the same calibration procedure; they are not competing claims that one optimizer is universally best. SC-FMA Ridge uses a closed-form softmax over a linear combination of $\tilde{\mathbf{c}}$ and $\tilde{\mathbf{n}}$. SC-FMA QP solves the full constrained program with redundancy and bottleneck terms. SC-FMA Projection applies a fast topology-constrained simplex projection onto the probability simplex [42]. The validation sections report where each variant is appropriate.

The calibrated allocation field is meant to be inspectable. For each selected artifact unit, the implementation can report whether the priority was driven mainly by fidelity, structural dependency, redundancy regularization, or bottleneck regularization.

This decomposition maps priorities to curation actions. A high-fidelity object may support a positive annotation signal; a high-dependency object may require manual review despite modest local utility; a redundant group may be audited as a cluster; and a bottleneck object may deserve conservative retention because later reasoning depends on it.

SC-FMA does not imply that the largest allocation value is the only important artifact in a sequence. The simplex constraint represents an audit budget, so increasing one unit's allocation value decreases another unit's share of attention. The output is therefore a fixed-budget audit record with explicit allocation fields. Redundant artifact units should not all receive high allocation values, while isolated downstream-exposed units should not disappear from review.

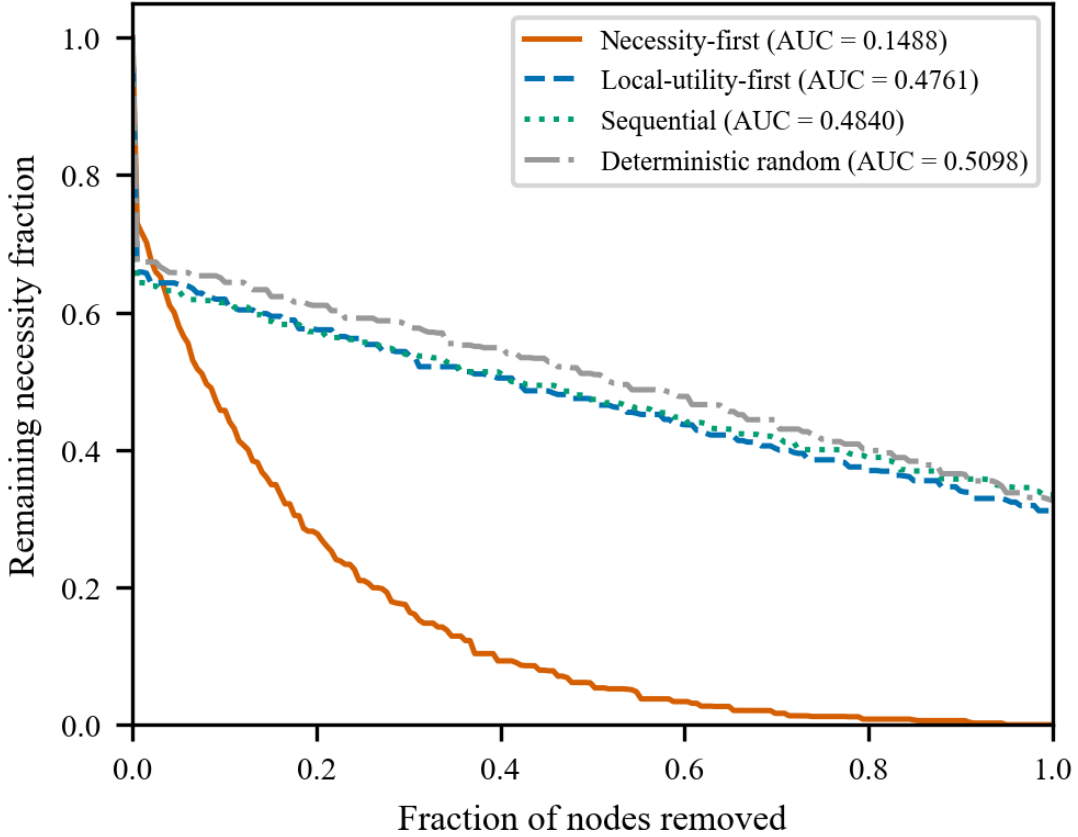


Figure 4: Resilience curves under ordered node removal. Four removal orders are reported: necessity-first, local-utility-first, sequential (original trace order), and deterministic random (hash-based). The y-axis shows remaining structural necessity fraction; the x-axis is the fraction of nodes removed. AUC values are annotated. The necessity-first curve degrades sharply (AUC 0.1488), while local-utility-first (AUC 0.4761) is close to random (AUC 0.5098), showing that structural criticality requires graph-level diagnostics beyond local utility.

3.7. Role of Structural Components

The structural components are not designed to exceed a strong fidelity signal in every setting. Their role is to preserve useful annotation signal when the fidelity field is strong and add audit information that a scalar-only view cannot expose: local support, structural exposure, redundancy, and bottleneck protection.

A fidelity-only scalar indicates which artifact receives attention but collapses the rationale. SC-FMA changes the interpretability of the audit record even when the allocation field changes only modestly: the contribution is preservation with decomposition.

Table 1

Asymptotic time complexity for SC-FMA audit-record construction. Here N is the number of artifact sequences, k is the number of auditable units in a sequence, L is text length, d is the sparse text-feature dimension, and I is the number of QP solver iterations.

Stage	Input size	Time complexity
Artifact feature extraction	N sequences, k units, text length L	$O(NkL)$
Graph construction	k nodes, sparse TF-IDF features d	$O(k + k^2d)$
Redundancy and bottleneck diagnostics	k units, similarity matrix	$O(k^2d)$
Ridge calibration	k artifact-feature rows	$O(k)$ after learned coefficients
QP calibration	k units, dense constraints	$O(Ik^3)$ worst case
Budgeted audit-record readout	k audit-record fields	$O(k \log k)$

Table 2

Memory profile for the same audit-record construction pipeline stages. Dense graph quantities are shown for clarity; the implementation uses sparse storage where applicable.

Stage	Memory complexity
Artifact feature extraction	$O(k + d)$ per sequence
Graph construction	$O(k^2)$ dense; sparse in implementation
Redundancy and bottleneck diagnostics	$O(k^2)$
Ridge calibration	$O(k)$
QP calibration	$O(k^2)$
Budgeted audit-record readout	$O(k)$

3.8. Computational Profile

Empirical runtime is reported separately from the asymptotic summaries. In the frozen local artifacts, the process-annotation audit-prioritization run over 4,417 traces and 34,219 labeled artifacts completed in 131.57 s. Separately logged graph-construction, SCU component, and failure-taxonomy runs are reported in Appendix B.

4. Knowledge Engineering Task Validation

This section instantiates fixed-budget knowledge audit as a knowledge-engineering and information-representation task. The experiments analyze representation behavior, information preservation, and audit fidelity rather than model-comparison performance; correlation, NDCG, and retrieval metrics are used only as diagnostic indicators for the resulting records. The validation suite examines three representation questions: whether audit records preserve useful annotation signals from a process-annotation dataset, whether graph construction changes structural-dependency representations, and whether decomposed records improve audit-target retrieval under budget constraints. Each stage follows the workflow in Figure 1: represent intermediate knowledge artifacts, construct an audit graph, calibrate an annotation or utility signal, and return an audit record. We begin by defining the fixed-budget audit task, then study process-annotation representation behavior, graph-based representation sensitivity, audit-record construction, controlled calibration, and failure modes of the representation layer.

4.1. Task Definition: Fixed-Budget Knowledge Audit

4.1.1. Data Sources and Reference Views

The process-annotation dataset stage uses PRM800K records under the frozen v3.6/v3.8 hash split (4,417 samples and 34,219 labeled artifacts). Here process supervision is treated as a source of artifact-level annotation signals, not as the paper's main contribution. The hash split fixes sample assignment before locked validation to avoid post hoc selection. The Countries-KG stage uses a real knowledge graph as a representation-backend study. The controlled synthetic stage contains 200 traces generated with five fixed seeds (42, 123, 456, 789, 1024), with approximately 1,000 reflective artifacts per seed, and is used for representation calibration, ablation, and error analysis. Controlled annotation

targets combine local correctness, lexical diversity, position, and graph-derived necessity. MuSiQue is reported as a supplementary constructed-label feasibility demonstration because its labels and feature fields share `step_type`; full metrics and execution details are reported in the supplementary material. WebQSP trace-audit artifacts are reported as a supplementary fixed-schema diagnostic on executable traces (494 train traces for development and 1,627 locked test traces). GSM8K/HotpotQA task-specific replay and downstream filtering attempts are retained as failed or incomplete validation attempts.

Reference views include raw local utility, span length, relative position, seeded random field assignment, and stage-specific scalar or feature-only views. Token-level attribution families such as gradient attribution, attention rollout, Monte Carlo Shapley, token surprisal, and step entropy are not computed under the archived workflow because the synthetic artifacts store per-step local utility, necessity, and redundancy rather than token-level activations; they are therefore discussed only under the reproducibility scope. The primary readout is annotation-order consistency for the audit-priority field, measured by Spearman ρ ; Kendall τ , record-level NDCG@3 and NDCG@5, and fixed-budget audit-coverage readouts are reported where relevant. Hyperparameter grids, including the gamma/delta and alpha/beta QP sensitivity grids (Supplementary Tables C.5 and C.6), additional ablations, error modes, and efficiency measurements are reported in the supplementary material.

The controlled synthetic stage is designed to test calibration behavior rather than to mimic all properties of natural reasoning. Local utility values are skewed, necessity is zero-inflated, and redundancy is block structured. These choices reflect the diagnostic pattern that motivated SC-FMA: local utility can be dense, while structural necessity is sparse. The controlled annotation target is deliberately not identical to any SCU input. It combines correctness, lexical diversity, position, and structural information so that the constructed audit field cannot be explained by copying one input vector. Supplementary Table D.12 reports cross-correlations between annotation-target components and SCU inputs to document this separation.

For the larger real-data stage, PRM800K is treated as a process-annotation distribution. The locked v3.6 split examines how well the record's fidelity and priority fields preserve annotation-order consistency under a fixed audit budget. The v3.8 frozen PRM reference provides in-distribution context using prefix-sequence reward probabilities. Because this public process-annotation dataset can overlap with public PRM training distributions, Section 5.4 handles the scope of PRM-training and best-of- N extensions.

All reported readouts are deterministic under the archived configs. The controlled synthetic stage uses five fixed seeds (42, 123, 456, 789, 1024) with variance reported as $\pm$ standard deviation. Supplementary Table B.1 reports bootstrap standard errors by trace size, Supplementary Table C.8 reports process-annotation confidence intervals for the main calibration variants, and Supplementary Table C.9 reports the extended fixed-budget audit readout. Extended Holm-Bonferroni, Friedman/Nemenyi, paired Wilcoxon, and effect-size analyses are retained in the supplementary reproducibility notes, while Section 5.4 centralizes scope limitations.

4.1.2. Validation Stages and Interpretation

The validation is staged because each dataset tests a different part of the knowledge-engineering workflow. Table 3 states the interpretation attached to each stage. This keeps process-annotation consistency, knowledge-graph representation, audit-target retrieval, controlled calibration, supplementary diagnostics, and failed replay attempts from being merged into a stronger claim than any individual stage permits.

This staged interpretation also governs null or negative results. The process-annotation variant readout shows that full QP can be too aggressive when the base fidelity field already aligns with annotation structure. The contribution is therefore not a predictive-superiority claim, but a reproducible way to combine local and structural evidence within stated data-generating conditions.

4.2. Process-Annotation Representation Behavior Study

The main real-data fidelity check uses PRM800K as a process-annotation distribution for annotation-signal preservation and fixed-budget audit prioritization. The stored $\mathbf{w}_{\text{struct}}$ field is the primary positive fidelity field, with Spearman $\rho = 0.611$ against PRM800K artifact annotations. Raw local utility is negative on this stage ($\rho = -0.077$), and the overlap-limited frozen PRM prefix-signal reference is lower ($\rho = 0.252$). SC-FMA Ridge closely preserves $\mathbf{w}_{\text{struct}}$ ($\rho = 0.604$), while QP ($\rho = 0.442$) and Projection ($\rho = -0.135$) are lower-consistency representation variants. Supplementary Tables C.8 and C.9 report the extended confidence intervals and fixed-budget readouts.

Table 4 shows that, on a process-annotation distribution, $\mathbf{w}_{\text{struct}}$ and Ridge retain more annotation-positive artifacts inside the fixed review prefix than raw local utility and simple reference views. The simple-average reference view

Table 3

Validation stages and interpretation. Each stage instantiates a separate part of the audit-prioritization workflow.

Validation stage	Primary artifact	Manuscript interpretation
PRM800K process-annotation stage	4,417 samples, 34,219 labeled artifacts	Process-annotation consistency and audit context.
Countries-KG diagnostic graph	Fixture-level Countries-KG diagnostic	Knowledge-graph backend and structural-priority sensitivity.
Audit-target retrieval stage	Rule-derived process-annotation audit targets	Automatic audit-target information recovery.
Controlled synthetic stage	200 traces, approximately 1,000 steps, five fixed seeds	Controlled calibration and ablation evidence.
Frozen PRM reference	Prefix-sequence reward signals	In-distribution reference context.
Supplementary MuSiQue/WebQSP diagnostics	Supplementary details only	Feasibility and fixed-schema diagnostic context only.
Failed GSM8K/HotpotQA attempts	Failed or blocked artifacts	Transparency only; no positive validation claim.

Table 4

PRM800K annotation-order consistency and fixed-budget audit coverage on the frozen split of 4,417 samples and 34,219 labeled steps. Metrics are displayed to three decimals from locked archived JSON artifacts and include all SC-FMA variants available in the locked package. Dashes for the Simple average row indicate that first-selected target and Mass@25% were not computed for this negatively correlated diagnostic reference view.

Field	Annot. ρ	First target	Mass@25%	Coverage NDCG@25%	Role
$\mathbf{w}_{\text{struct}}$	0.611	0.917	0.381	0.951	Primary fidelity signal.
SC-FMA Ridge	0.604	0.905	0.380	0.946	Preservation-with-decomposition approximation.
SC-FMA QP	0.442	0.903	0.379	0.944	Audit context on this stage.
SC-FMA Projection	-0.135	0.723	0.288	0.743	Lower-consistency projection variant.
Frozen PRM prefix signal	0.252	0.777	0.342	0.847	In-distribution reference context.
Raw local utility	-0.077	0.686	0.282	0.722	Direct ablation.
Simple average (local utility + necessity)	-0.443	–	–	0.438	Independent feature-average reference.
Random	0.006	0.733	0.307	0.776	Seeded sanity control.

(unweighted mean of local utility and structural necessity, Spearman $\rho = -0.443$) does not preserve annotation-order consistency, indicating that a calibrated representation field is needed when fidelity and structural signals are combined.

4.3. Graph-based Representation Sensitivity

The process-annotation stage studies annotation-order preservation in audit records, but not whether the graph interface can ingest domain relations. The Countries-KG representation study addresses that narrower question. As a diagnostic check on non-lexical graph construction, we use the Countries KG resource associated with approximate-reasoning evaluation over country, region, and neighborhood relations [43]. The fixture used here contains 30 entities and 189 triples over `locatedIn` and `neighborOf`. We generate 30 deterministic KG-query traces and construct both a lightweight lexical-dependency graph and a KG-augmented graph that maps ontology relations to functional edge types.

Table 5

Countries-KG knowledge-graph representation study (30 traces over a real knowledge graph, 30 entities, 189 triples). KG-augmented edges are derived from typed ontology relations.

Quantity	Value
KG entities / triples	30 / 189
KG relation types	locatedIn, neighborOf
Reasoning traces	30
TF-IDF reference edges	310
KG candidate edges	304
KG-augmented edges	320
KG-derived edges added	196
Ontology relations detected	depends_on_concept (211), same_constraint (85), neighbor_concept (8)
Mean abs structural-necessity delta	0.431
Structural-necessity consistency (reference vs KG)	0.0
Bottleneck top-1 overlap	1.000
Matched nodes	154

Table 6

Rule-derived audit-target retrieval on the locked PRM800K split. Failure-mode localization is evaluated as retrieval under a 25% budget. The SC-FMA condition uses decomposition fields; the scalar condition uses only w_{struct} . The targets are automatic rules rather than human adjudications, and their circularity levels are decomposed in Table 7.

View	Coverage@25%	Coverage NDCG@25%	MRR	First target	Cost Proxy
w_{struct} scalar only	0.235	0.353	0.354	0.000	0.0007
SC-FMA decomposition	0.699	0.978	1.000	1.000	0.0002

Table 5 shows that typed KG edges change the graph substantially: the stage considers 304 KG candidate relations, adds 196 ontology-derived edges, and yields 320 KG-augmented graph edges versus 310 TF-IDF reference edges. The mean absolute structural-necessity delta is 0.431 with a reference-vs-KG necessity consistency of 0.0, meaning KG augmentation substantially changes which objects the representation marks as structurally dependent rather than merely adding redundant connections. Full edge lists and the diagnostic report are in the supplementary material.

4.4. Audit Record Construction Validation

The audit-target retrieval stage evaluates whether decomposition fields recover rule-derived audit targets hidden by a scalar-only view. This stage is an automatic audit-record diagnostic, not independent human validation.

We evaluate failure-mode localization as retrieval over traces. The scalar-only view receives only the w_{struct} signal summary, while the SC-FMA view receives fidelity, necessity, redundancy, bottleneck status, and induced action fields. Automatic oracle targets are derived from frozen process-annotation labels, audit-priority disagreements, redundancy density, bottleneck flags, and the failure-taxonomy rules in Appendix A. This construction creates different circularity levels: weak utility-anchor targets have low direct overlap with the decomposition fields, whereas redundancy consolidation, bottleneck protection, and structural over-correction targets share partial circularity because their definitions use redundancy density, bottleneck flags, or structural disagreement fields also exposed by SC-FMA.

Table 6 reports the automatic retrieval readout. Table 7 separates the four target families so that low-circularity and partially circular targets are not merged into a single evidence claim.

The largest aggregate difference therefore should be read as recovery of rule-defined audit targets by a decomposed audit record, not as proof that SC-FMA independently predicts human audit choices. Table 8 gives one compact qualitative example. Full fixed-template Audit Cards are in Appendix A.

Table 7

Per-target audit-target retrieval under the same 25% budget. Circularity denotes overlap between the rule-derived target definition and SC-FMA decomposition fields. Recall and NDCG are copied from the locked auto-validation artifact.

Target family	Pos.	Circ.	w_{struct} R	SC-FMA R	w_{struct} N	SC-FMA N
Bottleneck protection	382	Partial	0.264	1.000	0.221	1.000
Redundancy consolidation	1,214	Partial	0.199	0.824	0.224	0.916
Weak utility anchor	2,420	Low	0.247	0.455	0.527	0.997
Structural over-correction	2,145	Partial	0.228	0.515	0.439	1.000

Table 8

Compact Audit Card case study: scalar-only view versus SC-FMA decomposition.

Failure target	w_{struct} -only view	SC-FMA Audit Card view
Bottleneck protection	A scalar signal can allocate a step for early inspection but does not identify that the selected step gates downstream reasoning.	Bottleneck and necessity fields mark the step as dependency-exposed, so the action is to protect or inspect the downstream chain before dropping it.
Redundancy consolidation	Multiple high-signal steps can look independently important.	Redundancy density and neighborhood fields identify a cluster, so the action is to consolidate the checks or sample the cluster rather than inspect every duplicate.
Weak utility anchor	A high calibrated fidelity field hides whether the raw local utility signal failed.	Fidelity and raw-anchor disagreement expose an upstream utility-anchor problem, so the action is to repair the signal source rather than blame graph structure.

Table 9

Controlled representation-field assignment on the synthetic stage (200 traces, approximately 1,000 steps). For SC-FMA variants, the Spearman ρ column is the 5-seed mean $\pm$ standard deviation (seeds 42, 123, 456, 789, 1024) and the remaining columns are from a representative seed (42), since the archived multiseed run records Spearman alone. All other reference views are from seed 42. Simple average is an independent reference view computed as the element-wise mean of normalized local utility and structural necessity. Token-level attribution families (gradient, attention-rollout, Shapley, surprisal, entropy) are not reported here because the archived synthetic pipeline stores only per-step local utility/necessity/redundancy rather than token-level activations, so those methods would collapse to raw local utility; they are discussed in the supplementary material under the reproducibility scope. Bootstrap confidence intervals and extended ablations are reported in the supplementary material.

View	Annot. ρ	Kendall τ	Record NDCG@3	Record NDCG@5
SC-FMA QP	0.597 $\pm$ 0.064	0.533	0.921	0.958
SC-FMA Ridge	0.541 $\pm$ 0.016	0.447	0.889	0.927
Raw local utility	0.483	0.412	0.874	0.919
Simple average (local utility + necessity)	0.478	0.413	0.889	0.926
SC-FMA Projection	0.372 $\pm$ 0.030	0.278	0.824	0.890
Span Length	-0.006	-0.013	0.740	0.826
Relative Position	-0.005	-0.004	0.735	0.827
Random	-0.010	-0.008	0.736	0.825

4.5. Controlled Calibration Analysis

Table 9 shows the intended calibration behavior on the controlled synthetic stage: SC-FMA QP records higher annotation-order consistency than raw local utility, with a Spearman ρ gap of 0.114 between the QP five-seed mean (0.597) and the raw local-utility representative-seed value (0.483). The readout supports the representation claim that structural calibration can repair local-utility allocation under controlled annotation targets. The next stage asks whether the graph interface can accept typed domain relations from a real knowledge graph.

Table 10

SCU component contribution on the controlled synthetic stage. Values are 5-seed mean $\pm$ standard deviation over seeds 42, 123, 456, 789, and 1024. Delta is the annotation-order consistency difference relative to `full_scu`. The largest drop occurs when the fidelity signal is removed; removing redundancy or bottleneck regularization has near-zero impact on the reported consistency readouts in this setting.

Variant	Annot. ρ mean $\pm$ std	Kendall τ mean $\pm$ std	Delta vs Full
<code>full_scu</code>	0.5103 $\pm$ 0.0320	0.4297 $\pm$ 0.0287	0.0000
<code>no_fidelity</code>	0.2850 $\pm$ 0.0153	0.2366 $\pm$ 0.0138	-0.2253
<code>no_structure</code>	0.4841 $\pm$ 0.0334	0.4053 $\pm$ 0.0285	-0.0263
<code>no_redundancy</code>	0.5093 $\pm$ 0.0321	0.4273 $\pm$ 0.0273	-0.0010
<code>no_bottleneck</code>	0.5090 $\pm$ 0.0304	0.4279 $\pm$ 0.0274	-0.0013

Table 11

SCU component contribution on a structural stress-test fixture (200 traces per seed, five fixed seeds 42, 123, 456, 789, 1024). Unlike the structure-independent labels of Table 10, these labels reward redundancy-awareness and bottleneck-protection under a stress-test redundancy setting ($\gamma = 0.6$). Delta is the annotation-order consistency difference relative to `full_scu`.

Variant	Annot. ρ mean $\pm$ std	Kendall τ mean $\pm$ std	Delta vs Full
<code>full_scu</code>	0.6249 $\pm$ 0.0145	0.5238 $\pm$ 0.0112	0.0000
<code>no_fidelity</code>	0.4832 $\pm$ 0.0357	0.3935 $\pm$ 0.0319	-0.1417
<code>no_structure</code>	0.5871 $\pm$ 0.0234	0.4870 $\pm$ 0.0224	-0.0378
<code>no_redundancy</code>	0.5758 $\pm$ 0.0163	0.4889 $\pm$ 0.0091	-0.0492
<code>no_bottleneck</code>	0.5409 $\pm$ 0.0145	0.4424 $\pm$ 0.0114	-0.0840

Table 10 refines the component narrative. The fidelity field is essential: removing it reduces annotation consistency by 0.2253, far more than any other removal. Graph necessity contributes moderately (Delta = -0.0263), while removing redundancy or bottleneck regularization changes annotation consistency by only about 0.001. These terms are therefore regularization and decomposition terms rather than representation-field drivers in this setting. The table reports `full_scu` Spearman 0.5103, lower than the SC-FMA QP value of 0.597 in Table 9. This is not a cross-table inconsistency: the two tables use different controlled synthetic configurations, and the within-table deltas are the meaningful quantities.

The near-zero redundancy and bottleneck deltas reflect a property of the labels, not of the terms. The proxy label in the original controlled stage is independent of redundancy and bottleneck structure, so the structural terms have nothing to align to. We therefore run a structural stress-test (Table 11) whose labels explicitly encode redundancy-awareness and bottleneck-protection. The fidelity signal blends this adjusted label with the pre-adjustment base label, remaining dominant but partially structurally blind.

On the stress-test, removing the fidelity signal still costs the most (-0.1417), but the structural terms now contribute materially: removing the bottleneck log-barrier costs 0.0840, the redundancy penalty 0.0492, and the structural-dependency term 0.0378. These values are far larger than the ≈ 0.001 redundancy and bottleneck deltas in the structure-independent setting. Redundancy and bottleneck are therefore *conditional* regularizers whose contribution scales with structural preference missing from the fidelity signal.

Table 12 shows that topical edges do not add annotation-signal preservation evidence in the diagnostic run: `full_tfidf` is slightly below `temporal_only` and matches `shuffled_topical` on the reported consistency readouts. The appropriate interpretation is therefore conservative. Topical edges can change structural diagnostics such as mean necessity, bridge-node fraction, and influence depth, but their observed role is supplementary structural cueing rather than a demonstrated representation-field driver.

The reference-view pattern clarifies the source of the representation signal. Raw local utility already provides more useful annotation-order information than span length, relative position, and random controls, so the supplied utility field carries relevant information. The independent simple average of local utility and structural necessity is comparable to

Table 12

Graph construction analysis on the diagnostic run. Temporal edges define the primary backbone; topical similarity edges are supplementary cues. Record NDCG@3 and Record NDCG@5 are interpreted as budget-constrained audit-coverage readouts computed from the same diagnostic-run source data and metric definition used for the correlation readout.

Variant	Annot. ρ	Kendall τ	Record NDCG@3	Record NDCG@5
full_tfidf	0.0515	0.0458	0.8655	0.8655
temporal_only	0.0596	0.0527	0.8688	0.8688
jaccard_topical	0.0517	0.0460	0.8657	0.8657
shuffled_topical	0.0515	0.0458	0.8655	0.8655

raw local utility; plain averaging does not preserve annotation order as well as calibrated SC-FMA fields. Token-level attribution families (gradient, attention rollout, Shapley, surprisal, entropy) are not reported because the archived synthetic pipeline stores per-step local utility/necessity/redundancy rather than token-level activations; those methods would collapse to raw local utility under the archived workflow.

The component and graph-construction analyses separate two claims. In this stage, SCU benefits primarily from fidelity-signal preservation and secondarily from graph-necessity alignment, while redundancy and bottleneck terms mainly support conservative decomposition. Separately, topical graph edges move structural diagnostics but do not yield an annotation-order consistency gain over temporal-only or shuffled-topical construction.

Error analysis identifies three main residual patterns. First, high-utility but zero-necessity steps can receive excessive audit allocation because the fidelity term still matters. Second, bottleneck nodes near the detector threshold can receive too little audit allocation when the indicator is not activated. Third, dense redundancy blocks can over-equalize allocation values when many steps share similar topical features.

4.6. Failure Modes of Representation Layer

The stratified audit-prioritization results diagnose when structural calibration helps or hinders process-annotation consistency. The multi-label failure taxonomy over 4,417 locked traces reports weak utility anchor in 54.79% of traces, structural over-correction in 48.56%, redundancy misclassification in 27.48%, low signal or tie in 18.32%, and bottleneck over-protection in 8.65%. Fixed-template Audit Cards are provided in Appendix A.

Trace length is the strongest observed predictor of variant behavior. On short traces (7,141 steps across 1,806 samples), $\mathbf{w}_{\text{struct}}$ reaches Spearman $\rho = 0.759$, Ridge preserves it at 0.757, and QP remains close at 0.734. On long traces (18,814 steps, 1,416 samples), $\mathbf{w}_{\text{struct}}$ drops to 0.447, Ridge to 0.430, and QP degrades to 0.172. Longer traces likely create denser redundancy blocks; QP's hard penalty then reallocates the audit-priority field away from the annotation signal, while Ridge's soft averaging avoids the sharpest overcorrection.

Label entropy gives a complementary view. On low-entropy traces (17,288 steps, 1,474 samples), discriminability is limited but $\mathbf{w}_{\text{struct}}$ and Ridge still record Coverage NDCG@25% above 0.99 as a fixed-budget audit-coverage readout. On high-entropy traces (5,983 steps, 1,304 samples), labels are more dispersed and QP ($\rho = 0.642$) moves closer to $\mathbf{w}_{\text{struct}}$ ($\rho = 0.705$), suggesting that explicit redundancy management is more useful when annotation structure is richer.

Across strata, $\mathbf{w}_{\text{struct}}$ provides the strongest fidelity reference, Ridge stays within 0.01–0.02 Spearman ρ across most strata, QP degrades on long or high-redundancy traces, and Projection provides the least consistent SC-FMA representation variant. The frozen PRM prefix signal provides moderate in-distribution reference context (Spearman 0.061–0.425) and remains consistently above raw local utility.

5. Discussion

5.1. Graph-Aware Calibration in the Knowledge Lifecycle

SC-FMA is best understood as a knowledge-engineering layer in the lifecycle from representation to organization, maintenance, evolution, and reuse. It does not replace existing reasoning systems; instead, it organizes their intermediate knowledge artifacts into auditable records for downstream maintenance and curation. Knowledge-intensive systems expose intermediate artifacts during construction, retrieval, validation, update, and reuse, but the exposed artifacts vary in their structural roles. Classic rule-based systems showed why rule chains and intermediate states must be inspectable [23, 25], while knowledge-engineering and curation models emphasize that represented knowledge must

remain maintainable over time [22, 41]. Recent KG prompting, engineering-design RAG, and KG-quality studies show why inspection remains difficult when structured knowledge is noisy, incomplete, redundant, or stale [35, 36, 38]. The lifecycle question is therefore how exposed evidence should be represented, organized, and prioritized under fixed-budget knowledge curation.

This is why the method is a knowledge representation contribution rather than a new reasoning algorithm or a new process reward model. SC-FMA reorganizes exposed artifacts into structured, maintainable records: a local fidelity field is retained, but the artifact is also located in a dependency graph, related to redundant neighbors, marked for bottleneck exposure, and linked to a maintenance action. The audit-priority field is one direct use of this representation; the broader contribution is the record schema that supports audit, update, reuse, and governance decisions.

The results suggest why graph-aware calibration is useful even when it does not exceed the strongest scalar fidelity signal. A scalar view can identify artifacts likely to be relevant, but it does not separate local support from structural dependency, redundancy, or bottleneck exposure. SC-FMA keeps these fields visible in the audit record, so the same audit-priority value can be interpreted as a curation decision: verify a dependency, consolidate duplicated evidence, protect a bottleneck, or repair a weak annotation anchor.

This distinction also addresses the post-processing alternative. A $\mathbf{w}_{\text{struct}}$ -plus-post-processing design could append explanatory labels after a scalar field has already fixed the record order, but it would not make fidelity, structural dependency, redundancy, bottleneck protection, and simplex budget consistency part of the same representation constraint. SCU calibrates those fields jointly: when $\mathbf{w}_{\text{struct}}$ is strong, Ridge behaves conservatively and preserves the supplied annotation-order field; when the supplied signal is weak or structurally blind, the calibrated field differs from naive feature addition, as shown by the negative simple-average reference in Table 4 and by the stress-test drops when bottleneck or redundancy terms are removed in Table 11. The value of SCU is therefore not only to exceed a strong scalar field, but to define a reproducible record-construction rule whose behavior changes when structure conflicts with the fidelity signal.

5.2. Audit Records for Knowledge Maintenance

Scalar priority alone answers “which artifact receives attention?” but not “what should the curator check?” A $\mathbf{w}_{\text{struct}}$ signal provides an effective fidelity field; SC-FMA adds whether the priority comes from fidelity, structural dependency, redundancy pressure, or bottleneck protection.

The process-annotation stratification explains why this helps audit prioritization even without a stronger scalar-only mechanism. Weak utility-anchor cases indicate when the fidelity signal should be repaired; structural-over-correction cases mark excessive graph regularization; redundancy cases identify clusters; and bottleneck cases identify dependency-exposed objects.

Budget-constrained audit coverage, first-selected target, and Mass@25% match this fixed-budget setting more directly than whole-trace correlation alone. In a knowledge-maintenance workflow, a curator may inspect only a small number of rule firings, retrieval checks, entity bindings, or reflective verification steps from each trace. Local evidence signals such as retrieval confidence, entity-linking similarity, and rule-certainty factors can duplicate earlier checks or miss dependency-exposed objects; SC-FMA records those distinctions.

5.3. Variant-Selection Guidance

Table 13 converts the stratified diagnosis into bounded development guidance. It should be calibrated on a development set before use, not treated as an optimal decision rule across new task distributions.

Ridge is the development choice when a fidelity signal already aligns with annotations, as in the locked process-annotation stage. QP should be evaluated when utility and structure are expected to disagree and redundancy management is central. Projection is computationally simple, but the current stratified evidence makes it brittle when structural signal can conflict with the fidelity signal.

5.4. Scope and Limitations

The current evidence supports audit-prioritization and representation-level knowledge-engineering claims. It does not validate a deployed knowledge-base maintenance workflow.

Limitation 1 (Controlled synthetic stage): The synthetic calibration stage is controlled and proxy-labeled (200 traces, approximately 1,000 steps per seed, with seeds 42, 123, 456, 789, and 1024). It supports calibration and ablation behavior under designed conditions, not natural-annotation validity.

Table 13

Ridge/QP selection guidance derived from the stratified analysis. This is an engineering guideline, not a theoretical guarantee.

Setting	Recommendation
Strong-fidelity signal, as in process-annotation distributions	Ridge
High structural conflict or weak-fidelity signal	QP, validated on development data first

Limitation 2 (stage-specific data bounds): The locked process-annotation stage supports annotation-order consistency and fixed-budget audit within a PRM800K-derived annotation distribution. Claims about other task distributions require separate validation, and SC-FMA Ridge is reported as preservation-with-decomposition rather than as exceeding w_{struct} .

Limitation 3 (Graph construction): The default graph uses temporal order and TF-IDF topical similarity. The Sentence-BERT ablation and Countries-KG typed-edge stage show backend substitutability, not deployed ontology-level validation.

Limitation 4 (Utility signal granularity): The binary-correctness local utility signal is a coarse trace-level utility anchor. The resulting audit-allocation values are local representation signals rather than causal effect estimates.

Limitation 5 (Downstream validation scope): The downstream-support evidence is limited to one rule-derived audit-target retrieval diagnostic. The targets are automatic rather than human-adjudicated, and redundancy consolidation, bottleneck protection, and structural over-correction targets are partially circular because their definitions use fields also exposed by SC-FMA. This stage therefore supports audit-record diagnostic coverage, not independent human audit quality, deployed maintenance impact, or broad downstream-task usefulness.

Scope exclusions. Four uses remain outside the current evidence:

- **Downstream PRM training:** SC-FMA has not been evaluated as a training signal for PRMs, reinforcement learning from process feedback, or best-of- N search.
- **Production knowledge-base validation:** The current validation does not include live deployment, domain-expert curation, downstream error reduction, or maintenance-cost measurement.
- **Causal identification:** SC-FMA estimates local functional attribution, not population-level treatment effects or counterfactual outcomes under unobserved confounding.
- **External PRM generalization:** The frozen PRM reference provides in-distribution context, with possible overlap between the process-annotation dataset and public PRM training distributions.

Within these bounds, SC-FMA points to a knowledge-lifecycle use case in which audit records support update decisions, redundancy consolidation, reuse of validated artifacts, and governance of exposed evidence. Stage-specific details, including GSM8K/HotpotQA replay failures, WebQSP KGQA boundaries, and QP-specific caveats, are documented in the supplementary material and appendices.

6. Conclusions

This paper proposed SC-FMA, a structurally calibrated representation layer that turns supplied utility or annotation-fidelity signals into inspectable audit records for knowledge-intensive artifacts. Its SCU objective combines fidelity, structural dependency, redundancy, and bottleneck terms so that priority is reported with its curation reason.

The evidence supports this role across complementary stages: process-annotation preservation of a strong fidelity field ($\rho = 0.604$ versus 0.611 ; Coverage NDCG@25% 0.946 versus 0.951), Countries-KG representation sensitivity, rule-derived audit-target retrieval beyond a scalar-only view with circularity bounded by target type (coverage@25% $0.235 \rightarrow 0.699$; Coverage NDCG@25% $0.353 \rightarrow 0.978$), and controlled synthetic calibration (Spearman $0.483 \rightarrow 0.597$).

For Information Sciences, the significance of SC-FMA lies in making intermediate knowledge artifacts representable, auditable, and maintainable rather than merely sortable. By converting reasoning traces, graph nodes,

and process annotations into audit records with explicit fidelity, structural dependency, redundancy, bottleneck, audit-reason, and maintenance-action fields, the method connects knowledge representation to system-level tasks of organization, curation, reuse, and governance. Within the observable-artifact and fixed-budget audit setting validated here, SC-FMA contributes an information-structuring layer for auditable knowledge processing under resource constraints.

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Declaration of Competing Interest

The authors declare that they have no conflicts of interest related to this work.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used OpenAI GPT-5 to improve language clarity and assist with LaTeX formatting checks. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Data Availability

The PRM800K process-annotation dataset, MuSiQue, and WebQSP are publicly available from their original sources. Derived locked-split reports, audit-prioritization artifacts, trace-audit diagnostics, validation configurations, and reproduction scripts will be deposited in an anonymous public repository for review and released with the final article. Full commands, local raw-data expectations, output paths, and scope notes are documented in the supplementary reproducibility checklist.

CRedit authorship contribution statement

Haoran Ma: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization. **Ningning Wang:** Methodology, Validation, Writing – Review & Editing, Supervision, Project administration, Funding acquisition.

A. Failure Taxonomy Appendix

This appendix provides diagnostic support for selecting variants in process-annotation-distribution audit prioritization. It stays within the real-data audit-prioritization boundary used in the main text.

Each Audit Card’s Decision Label maps to one of three canonical actions derived from the SCU decomposition and unavailable from the fidelity-only scalar: protect a bottleneck, consolidate a redundant cluster, or repair an upstream utility anchor.

Table 14

Rule-based failure taxonomy distribution on the locked PRM800K split (4,417 traces). Labels are multi-label, so percentages need not sum to 100%.

Label	Count	Percentage
structural_over_correction	2145	48.56%
redundancy_misclassification	1214	27.48%
bottleneck_over_protection	382	8.65%
weak_utility_anchor	2420	54.79%
low_signal_or_tie	809	18.32%

Redundancy misclassification and bottleneck over-protection appear often enough to inform variant selection. The following fixed-template Audit Cards link audit-record field changes to reviewer actions. They provide diagnostic explainability evidence, not a substitute for a full user study.

Table 15

Audit Card 1: bottleneck-protected structural over-correction.

Field	Audit Card
Trace	prm800k_pool_010077_26cf715105; 7 steps. Representative step: “Now, I wonder if there is a geometric interpretation of the problem.”
Observed issue	Labels include structural_over_correction and bottleneck_over_protection. Step 3 has a below-maximum label (0.5) but receives the largest QP allocation value (0.2493).
Audit-priority field behavior	w_{struct} keeps the high-label early vector-setup steps near the top (0.8966–0.9510) and gives the final false conclusion near-zero audit allocation (0.0027).
SC-FMA decomposition	Fidelity: w_{struct} keeps the high-label setup steps high. Necessity: graph exposure marks the reinterpretation as structurally visible. Redundancy: little consolidation is needed in this short trace. Bottleneck: QP protects the lower-label step, creating the audit disagreement.
Why not w_{struct} alone?	w_{struct} gives the expected annotation order but not the reason a lower-label step became structurally protected. SC-FMA exposes that the disagreement is bottleneck-driven, not generic prioritization behavior.
Reviewer Decision	Inspect the bottleneck-protected lower-label step before accepting the structural floor; check whether the geometric reinterpretation is a true downstream dependency or an over-protected detour.
Decision Label	PROTECT_BOTTLENECK
Scope note	Diagnostic process-annotation Audit Card only; it supports variant selection and reviewer triage, not task-success validation.

Table 16

Audit Card 2: redundancy over-merging in a long trace.

Field	Audit Card
Trace	prm800k_pool_012035_e6055a1cb2; 30 steps. Representative step: “One way to do that is to pick a random value of x and find the corresponding value of $y = f(x)$, then swap the coordinates.”
Observed issue	Labels include <code>structural_over_correction</code> , <code>redundancy_misclassification</code> , and <code>bottleneck_over_protection</code> . Redundancy density is high (0.5749), and QP collapses several early high-label setup steps to nearly zero.
Audit-priority field behavior	w_{struct} assigns consistently high signal values to the graph-sketch, asymptote, intercept, and checking steps (roughly 0.91–0.97), while QP reallocates mass away from several of these high-label setup steps.
SC-FMA decomposition	Fidelity: w_{struct} keeps many graph-analysis checks high. Necessity: downstream exposure concentrates attention on a narrow band. Redundancy: similar graph checks are treated as overlapping. Bottleneck: protected steps compete with high-label setup steps for review budget.
Why not w_{struct} alone?	The scalar signal shows which steps are high, but it cannot tell whether the trace contains redundant audit units or whether several superficially similar graph checks should be consolidated. The decomposition makes that audit decision explicit.
Reviewer Decision	Check redundancy over-merging: decide whether asymptote identification, intercept checks, and coordinate-swap verification are separate audit units or can be reviewed as one cluster.
Decision Label	CONSOLIDATE_REDUNDANT_CLUSTER
Scope note	Diagnostic process-annotation Audit Card only; it motivates reviewer workflow design and does not validate live knowledge-base traces.

Table 17

Audit Card 3: weak local-utility anchor.

Field	Audit Card
Trace	prm800k_pool_006217_43f4f279ef; 3 steps. Representative step: “I call this altitude $\overline{AF}$, where F is the midpoint of BC .”
Observed issue	Labels are [0.5, 1.0, 0.0]. Raw local utility is inverted: it gives the false parallel-line step the largest signal value (1.2159) and the highest-label midpoint-altitude step a lower value (0.7647).
Audit-priority field behavior	w_{struct} , QP, and Ridge allocate the midpoint-altitude step for early audit and the false parallel-line step for late audit; raw local utility has Spearman $\rho = -1.0$ on this trace.
SC-FMA decomposition	Fidelity: the supplied fidelity field corrects the raw utility inversion. Necessity: the altitude construction remains the structurally meaningful check. Redundancy: no consolidation decision is needed in the three-step trace. Bottleneck: no bottleneck flag is needed, so the reviewer should repair the utility anchor rather than the structure detector.
Why not w_{struct} alone?	w_{struct} fixes the local allocation pattern, but the decomposition identifies the failure mode as a weak utility anchor rather than redundancy or bottleneck pressure. That distinction tells the reviewer which upstream signal needs repair.
Reviewer Decision	Inspect why the local utility anchor rewarded the false parallel-line step; use the structured decomposition for triage and avoid treating raw utility as an independent audit target.
Decision Label	REPAIR_UPSTREAM_UTILITY
Scope note	Diagnostic process-annotation Audit Card only; it supports signal-quality auditing, not external reasoning-task generalization.

B. Runtime and Reproducibility

Runtime and reproducibility metadata are summarized in Table 18. The records below are wall-clock measurements from the reproducibility summary; API calls remain zero and frozen artifacts are read-only inputs.

Table 18

Runtime and reproducibility summary for the validation runs. Wall time is measured in seconds; API calls are zero for all runs.

Route	Samples	Steps	Seeds	Elapsed (s)	Hardware	API Calls
SCU Component Contribution	200	1,008	5	30.7961	CPU (12 cores)	zero
Graph Construction Ablation	800	2,400	5	3.4197	CPU (12 cores)	zero
Failure Taxonomy	4,417	34,219	N/A	88.6221	CPU (12 cores)	zero

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